

Attempt all questions

1. Answer any 4 from the following questions

- What is Big O notation? Why do we use Big O notation to compare algorithms?
- Calculate the address of a random element present in a 2D array of size $M \times N$, given the base address as *baseArr*.
- Why are recursive subroutines not encouraged in time-bound / time-critical applications?
- Explain the scenarios when you prefer to use linear search over binary search.
- "Dangling pointer can cause memory corruption" – justify.
- What is a stack overflow error, and when it happens?

(4x2.5 = 10)

2. Answer any 2 from the following questions

- What are the limitations of array based implementation of a linear queue? How these limitations can it be removed?
- Write down the recursive subroutine (using C programming language) for calculating the resulting number for an input positive number n for the following definition of recursive function $f(n)$. Then trace the function calls for $n=4$.
$$f(1) = 2;$$
$$f(2) = 4;$$
$$f(n) = f(n-1) + n * f(n-2);$$

c) Explain why complexity of Binary Search is $O(\log n)$?

(2x4 = 8)

3. Answer any 2 from the following questions using C programming language

- Suppose L_1 and L_2 are two sorted singly linked lists of integers. Write down a procedure to merge them together to create another singly sorted linked list L_3 .
- Define the node structure for a queue using linked node representation. Develop the queue subroutines `insertQueue(...)` and `deleteQueue(...)`.
- A given string consisting of the characters a and b with other characters. For example, the string, "abbbccaaba" has equal number of a 's and b 's, whereas the string "acbbaabbcbb" does not. Write a subroutine to confirm that the string has equal numbers of a 's and b 's using stack (*no need to write stack related functions*).

(2x6 = 12)